

HOMEWORK 1

DUE: OCTOBER 7

Do at least 10 of the following problems.

EXERCISES FROM BONNAFÉ

1.2, 1.9, 1.12, 1.14, 1.15, 2.2, 2.3, 2.5, 3.2, 3.4

EXERCISES NOT FROM BONNAFÉ

- (1) Let X be a scheme and let Γ be a finite group acting on X by scheme automorphisms (on the right). Let $Y = X/\Gamma$ as a topological space, let $\pi: X \rightarrow Y$ be the projection (as topological spaces) and let $\mathcal{O}_Y = (\pi_* \mathcal{O}_X)^\Gamma$.
 - (a) Show that $\mathcal{O}_Y$ is a sheaf on Y .
 - (b) Assume that for every $x \in X$, there is an open affine $U \subseteq X$ such that $\Gamma x \subseteq U$. Prove that $(Y, \mathcal{O}_Y)$ is a scheme and that $\pi: X \rightarrow Y$ is a quotient of X by Γ .
 - (c) Show that if X is quasiprojective over a field, then the condition in part (b) is satisfied.
- (2) Prove the tensor-hom adjunction we used in class. More specifically, let R and S be rings and let M be an (R, S) -bimodule. Prove that the functor $S\text{-Mod} \rightarrow R\text{-Mod}$, $V \mapsto M \otimes_S V$ is left adjoint to the functor $R\text{-Mod} \rightarrow S\text{-Mod}$, $W \mapsto \text{Hom}_R(M, W)$.
- (3) Let Γ be a finite group. For a commutative ring R , we let $R\Gamma$ be the group algebra of Γ over R . Let M be a $\mathbb{Z}\Gamma$ -module. Let M^Γ be the invariants of Γ in M (i.e. the maximal Γ -invariant submodule) and let M_Γ be the Γ coinvariants of M (i.e. the maximal Γ -invariant quotient).
 - (a) Prove that if multiplication by $|\Gamma|$ is an isomorphism of M , then the natural map $M^\Gamma \rightarrow M_\Gamma$ is an isomorphism.
 - (b) Use part (a) to give a proof of the following fact used in Lecture on September 18: Let K be a field whose characteristic does not divide $|\Gamma|$. Prove that for any $K\Gamma$ -modules M and N , finite over K , there is an isomorphism $\text{Hom}_{K\Gamma}(M, N) \cong \text{Hom}_K(M, K) \otimes_{K\Gamma} N$.
- (4) Take any proof from lecture that you either didn't completely understand or in which details were skipped (and not covered by exercises above) and fill in the details completely to your satisfaction.